In the Name of Allah

Signals and Systems Final Project
Audio Separation Using a Notch Filter

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Part 1

For the `Fourier_Samples` script, utilizing the Fourier series approximation for the N samples obtained from the audio file, we calculate the Fourier transform of $x(t)$ at each specific frequency point requested in `freq_vector`.

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \tag{1}$$

$$X(2\pi f_i) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi f_i t}dt \tag{2}$$

Since $t \approx kT_s = \frac{k}{F_s}$, it implies that $dt \approx \frac{1}{F_s}$. Thus:

$$X(2\pi f_i) \approx \frac{1}{F_s}\sum_{k=0}^{N-1} x\left(\frac{k}{F_s}\right)e^{-j2\pi f_i \frac{k}{F_s}} \tag{3}$$

Let $x[n] = x\left(\frac{n}{F_s}\right)$, therefore:

$$X(2\pi f_i) \approx \frac{1}{F_s}\sum_{k=0}^{N-1} x[k]e^{-j2\pi f_i \frac{k}{F_s}} \tag{4}$$

Part 2

In the frequency spectrum, the human voice exhibits a smooth and broad profile, whereas the siren sound is distinctly sharper and narrower. By examining the audio file's frequency spectrum, one can identify the sharp, prominent frequencies and use them to determine the notch frequencies for our filter.

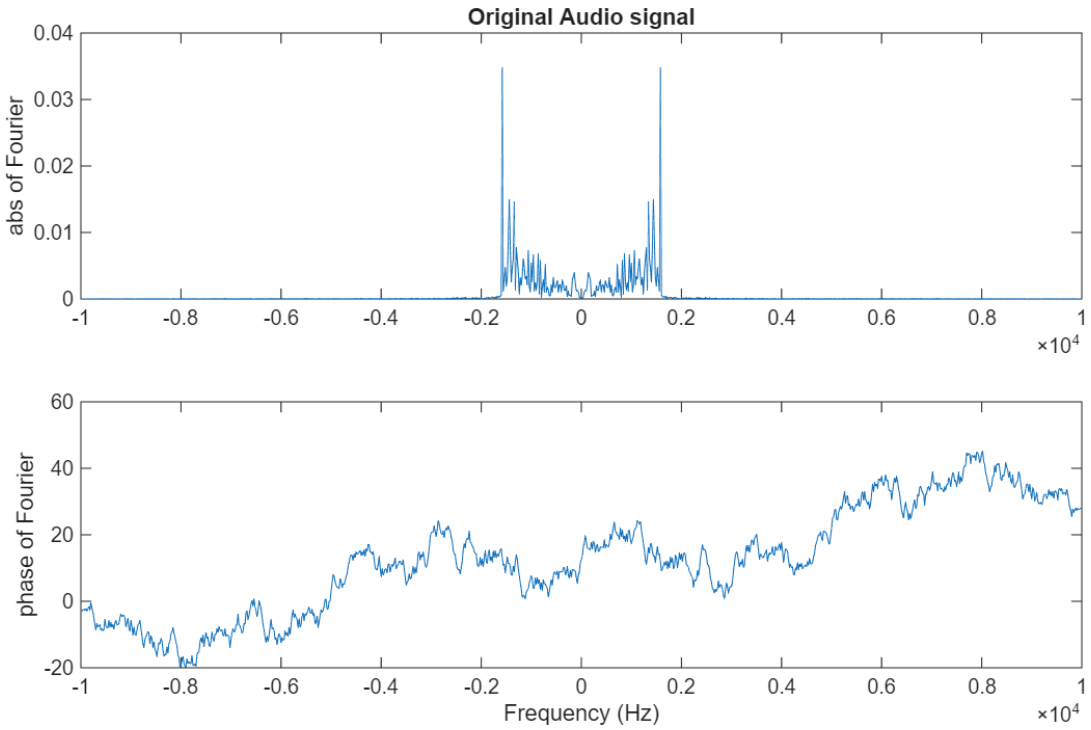


Figure 1: Fourier Transform of Original Audio Signal

The filter design strategy involves placing zeros on the unit circle exactly at the specific frequencies extracted from the siren sound. In close proximity to these zeros—for instance, at $r = 0.99$ and at the same angle—a pole is placed so that the filter highly attenuates signals near that frequency while leaving distant frequencies unaffected.

Accordingly, we select the 8 sharpest and highest-magnitude frequencies from the input signal's Fourier transform magnitude plot and place them on the z-plane using the aforementioned method.

$$\sum_{k=0}^{N_y} a_k y[n-k] = \sum_{k=0}^{N_x} b_k x[n-k] \tag{5}$$

$$\Rightarrow \sum_{k=0}^{N_y} a_k Y(z) z^{-k} = \sum_{k=0}^{N_x} b_k X(z) z^{-k} \tag{6}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{N_x} b_k z^{-k}}{\sum_{k=0}^{N_y} a_k z^{-k}} \tag{7}$$

For arbitrary frequencies such as ω_1 and ω_2:

$$H(z) = \frac{(1 - e^{j\omega_1} z^{-1})(1 - e^{-j\omega_1} z^{-1})(1 - e^{j\omega_2} z^{-1})(1 - e^{-j\omega_2} z^{-1})}{(1 - re^{j\omega_1} z^{-1})(1 - re^{-j\omega_1} z^{-1})(1 - re^{j\omega_2} z^{-1})(1 - re^{-j\omega_2} z^{-1})} \tag{8}$$

$$H(z) = \frac{(1 - 2\cos\omega_1 z^{-1} + z^{-2})(1 - 2\cos\omega_2 z^{-1} + z^{-2})}{(1 - 2r\cos\omega_1 z^{-1} + r^2 z^{-2})(1 - 2r\cos\omega_2 z^{-1} + r^2 z^{-2})} \tag{9}$$

$$H(z) = \frac{1 - 2(\cos\omega_1 + \cos\omega_2) z^{-1} + 2(1 + 2\cos\omega_1 \cos\omega_2) z^{-2} - 2(\cos\omega_1 + \cos\omega_2) z^{-3} + z^{-4}}{1 - 2r(\cos\omega_1 + \cos\omega_2) z^{-1} + 2r^2(1 + 2\cos\omega_1 \cos\omega_2) z^{-2} - 2r^3(\cos\omega_1 + \cos\omega_2) z^{-3} + r^4 z^{-4}} \tag{10}$$

From this resulting equation, the elements of `b_vec` and `a_vec` can be extracted.

Regarding the practical implementation of the filter in MATLAB, to achieve greater stability and precision, four 4th-order filters (with the transfer function described above) were utilized. These four filters were cascaded in series so that their transfer functions multiply, yielding the targeted 16 zeros and poles. Furthermore, for the higher frequencies in the file, $r = 0.95$ was used to provide broader attenuation around the primary siren frequencies (since the human voice has relatively less energy in that range). Conversely, $r = 0.975$ was used for the lower frequencies. Essentially, the lower frequencies correspond to the lower formants, which contain more of the speaker's vocal energy, making their preservation more critical.

To achieve further attenuation of the siren sound, the original audio signal is passed through the complete filter twice. In essence, the entire filter block is applied in a cascaded configuration.

Finally, the magnitude and phase plots of the output audio's Fourier transform are presented below. What is evident in the output audio is a slight distortion and muffling of the speaker's voice—due to the elimination of some of its frequency components—coupled with a highly effective attenuation of the siren sound. While further attenuation of the siren is possible, doing so would cause the police officer's voice to become overly distorted and hollow.

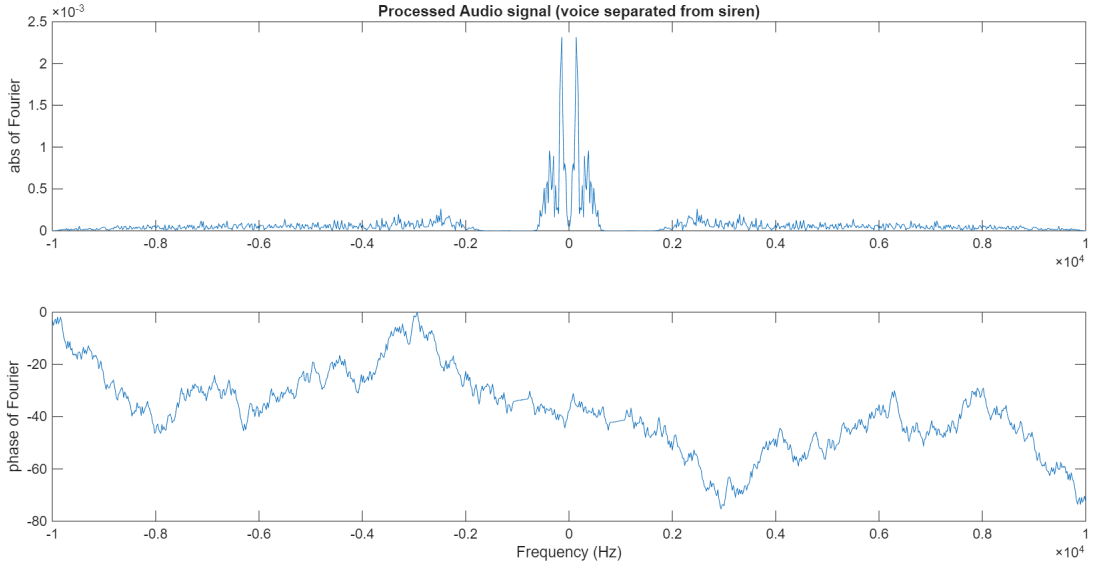


Figure 2: Fourier Transform of Processed Audio Signal